

Characterizing Transport Properties of Surface Charged Nanofiltration Membranes via Model-Based Data Analytics

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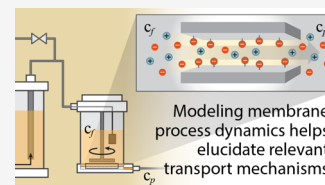


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ABSTRACT: This study presents an integrated framework for characterizing the transport properties of surface-charged nanofiltration (NF) membranes through dynamic diafiltration experiments and model-based data analytics. By incorporating startup dynamics, time corrections, and the influence of water flux on solute permeation, the framework effectively captures key transport properties governing membrane performance. A comparison of lag and overflow experimental modes highlights the superior precision of the lag startup mode, improving parameter estimates by 8%, 138%, and 83% as quantified by A-, D-, and E-optimality, respectively. Diafiltration experiments in the diluting regime further enhance model discrimination, enabling the identification of dominant transport mechanisms. This work marks a step toward developing self-driving laboratories (SDLs) that leverage model-based design of experiments (MBDoe) to expedite the characterization and development of high-performance NF membranes. The framework provides critical insights into transport phenomena, enabling the inverse design of membranes tailored to the demands of modern separation systems.



INTRODUCTION

Membrane technologies can enhance the sustainability and energy efficiency of modern separation processes.^{1,2} Among these, nanofiltration (NF), a pressure-driven membrane separation process, occupies a niche between ultrafiltration (UF) and reverse osmosis (RO), offering unique selectivity for multivalent ions,³ organic molecules,⁴ and small solutes.⁵ Unlike UF, which primarily separates based on size, and RO, which can reject most solutes including monovalent ions, NF membranes provide selective separation through a combination of size and charge effects. For example, the surface charge of NF membranes plays a crucial role in solute selectivity, allowing them to reject charged species based on electrostatic interactions.^{6–9} With pore sizes on the order of 1–3 nm and the ability to exploit both size exclusion and charge-based mechanisms, NF membranes are effective for applications such as water hardness reduction,¹⁰ contaminant removal,¹¹ and the recovery of critical materials^{12,13} from industrial streams. This versatility makes NF applicable in diverse fields, including wastewater treatment, chemical separations, and environmental remediation. The combination of relatively low energy consumption, size selectivity, and charge-based rejection positions NF as a valuable and adaptable technology in addressing global grand challenges such as equitable access to clean water, affordable sustainable energy, and circular economies.¹⁴

Broader applications of NF membranes are constrained by trade-offs between their permeability and selectivity.¹⁵ Recent studies suggest that material innovations have led thin-film composite (TFC) membranes to approach the upper bound of water permeability.^{16,17} Thus, selectivity has become an increasingly important criterion when screening membranes for practical applications, such as in the recovery of rare earth elements.¹⁸ Ongoing efforts to enhance membrane selectivity have primarily focused on two directions: the precise control of membrane nanostructure and the identification of chemical functionality that promotes the desired transport properties.^{19,20} Tailoring membrane surfaces to interact with specific solutes (e.g., through electrostatic interactions, hydrophobicity, or specific binding sites) shows promise in controlling solute transport properties.^{21–25} However, current research in this field remains largely exploratory (i.e., driven by trial-and-error approaches) and relies on traditional, human-driven experimental approaches. While effective in certain contexts, these approaches rely on simplified analysis, which often fails to capture the intricate interplay of factors that influence

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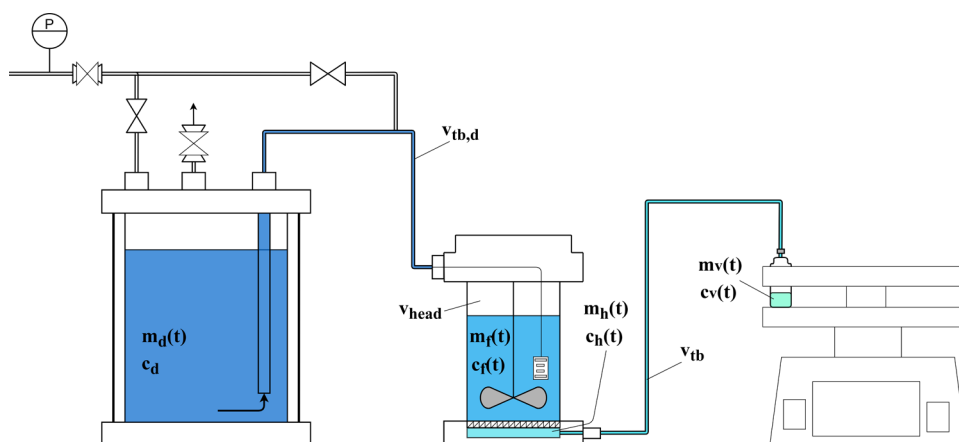


Figure 1. Pressure is first applied to both the diafiltrate tank and the head space of the stirred cell to bring the whole system to the initial pressure. The valve to the stirred cell is then closed and an increased pressure is applied to the diafiltrate tank (subscript d). The diafiltrate flows through the tubing into the stirred cell containing the feed solution (subscript f), with concentration c_f monitored by an inline conductivity probe. The permeate initially enters the holdup region (subscript h) and then travels through a tube with volume, v_{tb} , before being collected in scintillation vials resting on a balance. Time-series mass data, m_v , are recorded, and the average permeate concentration, c_v , is analyzed using inductively coupled plasma-optical emission spectroscopy (ICP-OES) Reproduced from Muetzel et al.⁵⁴ Copyright 2022 American Chemical Society.

transmembrane transport.²⁶ As one example, the effect of solute-membrane interactions on diffusion and partitioning remains largely convoluted due to the lack of information contained within measurements. Traditionally, the experiments used to regress these coefficients rely on ex-situ solution analyses (e.g., sorption²⁷ and time-lag experiments,²⁸ respectively) and have changed little over the past 40 years. The statistical design of experiments (e.g., response surface methodology) provides a systematic approach to planning experiments and identifying factors that influence membrane performance, but requires time-consuming and resource-intensive data collection before valid conclusions can be drawn.^{29,30} Consequently, there is a pressing need for more robust, adaptable experimentation methods, as well as sophisticated modeling techniques, to accurately characterize and deconvolute transport properties under a wide range of conditions.

Recent advancements in automation and artificial intelligence (AI) have introduced transformative approaches to material research through the development of self-driving laboratories (SDLs) and digital twins.^{31–34} SDLs integrate experimental automation with AI-driven optimization, enabling the autonomous execution of experiments, data collection, analysis, and iterative refinement of experimental conditions. While SDLs can revolutionize experimentation, they typically rely on classical “black-box” approaches that fail to integrate mechanistic understanding,^{31,35,36} limiting their ability to generalize findings across complex membrane systems. Digital twins complement SDLs by providing virtual representations of physical systems that enable predictive modeling, real-time system monitoring, and data-driven decision-making.^{37,38} This combination further enhanced the efficiency of SDLs in generating and analyzing experimental data.^{39–42} For NF applications, digital twins leveraging dynamic membrane characterization could simulate complex transport phenomena, predict membrane performance, and optimize separation process conditions.¹³ These tools create a valuable bridge between theoretical predictions and experimental validation. However, before future research efforts in the membrane field can pair SDLs and digital twins, accurate physics-based models and comprehensive data sets must be available.

To create comprehensive data sets, innovations that can inform the design of experiments to generate mechanistic insights are needed.³⁴ Model-based design of experiments (MBDoE) utilizes high-fidelity models grounded in scientific principles to optimize experimental campaigns. MBDoE enables the discrimination between scientific hypotheses^{43,44}—posed as mathematical models—and improves the precision of parameter estimates.^{45–47} For instance, MBDoE has been used to differentiate between candidate kinetic models in microreactor designs, increasing the information gained from experiments while reducing material costs and experimental time commitments.⁴⁸ While MBDoE has been employed to understand many processes (e.g., oxidation of methanol on silver catalysts,⁴⁹ product formulation⁵⁰), only a few of those applications (e.g., electrodialysis⁵¹ and chromatographic separations⁵²) come from the separations community. The emergence of new MBDoE software⁵³ holds significant potential for optimizing the design of instruments and experiments. Building on these advancements, parameter precision criteria from MBDoE have been applied to diafiltration experiments^{26,54,55} to optimize time-invariant experimental conditions in the system.⁵⁶ Applying MBDoE enabled a quantitative comparison between traditional filtration and dynamic diafiltration experiments. The results demonstrated that diafiltration experiments enhanced parameter estimation and reduced the experimental time required to characterize the transport properties of TFC NF membranes. By harnessing the power of MBDoE, researchers can systematically optimize experiments to maximize the information gained, reducing the time and resources needed for membrane development. The integration of data science techniques,⁴⁷ science-based digital twins, and SDLs into membrane research creates a powerful pathway toward more efficient experimentation, inverse design of materials, and the real-time optimization of processes.

In this work, we combine dynamic diafiltration experiments with model-based data analytics techniques to systematically characterize the transport properties of charged NF membranes (i.e., NF270 membranes, which exhibit a negative surface charge for aqueous solutions above pH 3). By incorporating experimental startup dynamics and accounting for the delay between when the permeate exits the membrane and when it is

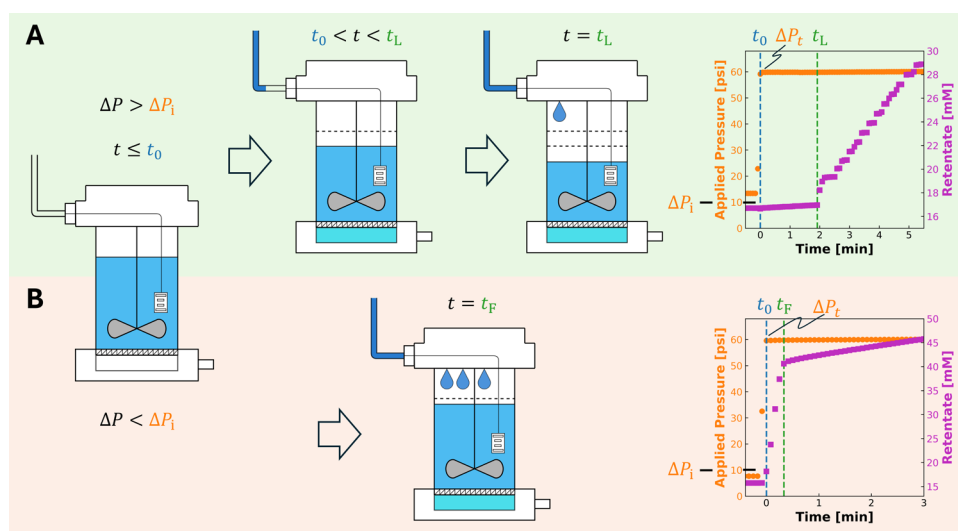


Figure 2. Two startup modes in which diafiltration can be operated. (A) In lag mode, the diafiltrate does not enter the stirred cell immediately upon reaching operating pressure ΔP_i at t_0 . Rather, the system acts as a filtration experiment for a period, $t_0 \rightarrow t_L$. During this time, the solution begins to permeate and slowly fill the holdup region. After t_L , drops of diafiltrate enter the stirred cell as drops of solution continue to permeate. (B) In overflow mode, a bolus of diafiltrate flushes into the stirred cell once the operating pressure has been reached. The time at which the diafiltrate bolus is detected by a spike in conductivity is denoted as t_F .

collected in scintillation vials, we develop a robust framework that accurately captures both process and membrane parameters. Namely, it was previously proposed that the experiments could be started in two distinct ways⁵⁴: experiments can be engineered such that (1) a lag occurs before diafiltrate is introduced into the stirred cell or (2) an overflow of diafiltrate enters the stirred cell at the beginning of the experiment. Here we identify which startup mode provides the largest amount of information. We also explore diafiltration experiments conducted in the diluting regime and identify that these experiments are uniquely able to discriminate between transport models. Importantly, the contributions of the diffusion coefficient and the partition coefficient to transport can be deconvoluted using the diafiltration system. Using these insights in advancing membrane characterization, we establish a pathway toward integrating mathematical modeling with experimentation for the inverse design of membrane materials, offering a data-driven approach to optimize membrane properties for various industrial and environmental applications. This study lays the groundwork for future advancements in membrane technology, providing scalable and efficient solutions for process optimization and the development of next-generation membrane-based separation systems.

MATHEMATICAL MODELING

The selective transport of ions across the NF270 membranes is influenced by a combination of diffusion, convection, and electromigration, which are driven by gradients in concentration, pressure, and electric potential, respectively.⁵⁷ To accurately characterize the relevant transport mechanisms, we employed a model-based data analytics approach that integrates mathematical models with experimental data to provide a comprehensive understanding of membrane behavior under varying operating conditions (i.e., feed-side or retentate concentration and pressure).

To investigate these transport mechanisms, dynamic diafiltration experiments^{26,54} were conducted. Figure 1 presents the automated diafiltration apparatus, where a diafiltrate

(variables denoted with a subscript d) is continuously pumped under the action of applied pressure, ΔP , into a stirred cell holding the feed solution (subscript f). The permeate product directly under the membrane accumulates in the holdup region (subscript h), after which it is directed through a tube (subscript tb) to a series of scintillation vials (subscript v) for collection and further analysis. The mass of the sample vial, m_v , and the concentration of the retentate in the stirred cell, c_p , are measured at a rate of 0.2 Hz. The permeate is collected in 1 g aliquots and the solution concentration in these scintillation vials, c_v , is measured after the experiment has concluded.

The system dynamics are governed by the volumetric flux of water, J_w , and the molar flux of the solute, J_s . In this study, diafiltration experiments were conducted at constant applied pressures with steadily changing feed concentrations, examining both concentrating and diluting regimes,⁵⁸ under the assumption of a well-mixed feed solution. In the concentrating regime, the retentate concentration in the stirred cell monotonically increases as the solute flux into the stirred cell exceeds the solute flux exiting in the permeate. In the diluting regime, the opposite occurs, where the solute flux entering the stirred cell is lower than the solute flux leaving with the permeate, resulting in a dilution of the retentate concentration over time.

Previous work^{26,54} (i.e., the model used in the DATA framework) assumed that the solution volume in the stirred cell remains constant over the course of the diafiltration experiment (i.e., the rate of diafiltrate entering the stirred cell is equal to the volumetric flow rate of the leaving permeate). However, careful visual observations of the apparatus reveal changes in the feed solution volume within the stirred cell during startup processes, affecting the model initialization and subsequent predictions. The volume changes, occurring both during startup and throughout the experiment, introduce deviations from the initial equations of the diafiltration model, thereby affecting the accuracy of its predictions. In this work, we show that modeling the system startup dynamics increases the information gained from diafiltration experiments.

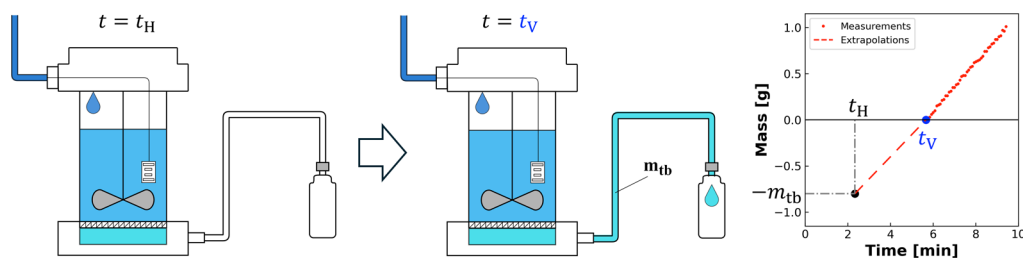


Figure 3. Time corrections considered for the permeate solution collected in scintillation vials. The diafiltration apparatus is subject to a delay in the time associated with the collection of permeate in the scintillation vial due to the tube filling with mass m_{tb} . $t_V - t_H$ represents the time delay for the first drop of permeate collected in the first vial. Accounting for this delay allows the permeate and retentate solution concentrations to be correlated correctly over the course of the experiment.

Startup Modes: Lag and Overflow. Ideally, during startup, the increase in pressure at t_0 brings the diafiltrate exactly to the entrance of the stirred cell. To achieve this, the air in the tubing between the diafiltrate tank and the stirred cell ($v_{tb,d} \sim 35$ mL) and the air in the head space of the stirred cell ($v_{head} \sim 7$ mL) needs to be compressed into the head of the stirred cell. If the experimental operating pressure is defined and it is assumed that the gas behaves ideally, the initial pressure that the tubing must be pressurized to (ΔP_i) can be calculated from the desired operating pressure (ΔP_t) in eq 1⁵⁴:

$$\Delta P_i = \frac{v_{head}}{v_{head} + v_{tb,d}} \Delta P_t \approx \frac{7}{7 + 35} \Delta P_t = \frac{1}{6} \Delta P_t \quad (1)$$

Pressurization of the tubing between the diafiltrate tank and stirred cell is accomplished through a series of valves. After the initial pressurization of the tubing, the diafiltrate tank is pressurized to the operating pressure ΔP_t . The moment the applied pressure reaches the desired operating pressure, the diafiltration experiment has begun and the dynamic modeling begins, marked as time t_0 . In practice, hitting the initial and operating pressure precisely is difficult. As such, experiments are run in either lag mode or overflow mode as depicted in Figure 2.

In the lag mode, the initial system pressure is too large and the desired operating pressure is not sufficient to bring the diafiltrate to the entrance of the stirred cell. As such, this condition initially emulates a filtration experiment: solution permeates through the membrane, leading to a decrease in feed volume and an increase in the feed concentration within the stirred cell. Therefore, for $t_0 \leq t < t_L$, the mass of the diafiltrate, m_d , remains constant, while the mass of the feed solution, m_f , decreases due to the transmembrane flux, expressed by eqs 2 and 3:

$$\frac{dm_d}{dt} = 0 \quad (2)$$

$$\frac{dm_f}{dt} = -\frac{dm_d}{dt} - A_m \rho J_w = -A_m \rho J_w \quad (3)$$

where A_m is the membrane area and ρ is the density of the solution. As the volume of the solution in the stirred cell decreases due to permeation, the volume of the air in the tube decreases. Once the air in the tube is incorporated into the stirred cell, the first drop of diafiltrate enters the stirred cell, marking the end of the lag phase at time t_L . The subsequent dripping of diafiltrate into the stirred cell causes a steeper change in the feed concentration, c_f .

In the overflow mode, the initial system pressure is too small and a bolus of diafiltrate immediately enters the stirred cell. This overflow of diafiltrate causes a sharp change in the feed concentration. The moment the last drop of the bolus enters the

stirred cell marks the end of the overflow phase at time t_F . In overflow mode, for $t_0 \leq t < t_F$, it is assumed that the diafiltrate bolus leaves the tube and enters the stirred cell at a constant rate $\dot{V}_o$, while permeation also occurs, represented by eqs 4 and 5:

$$\frac{dm_d}{dt} = \dot{V}_o \quad (4)$$

$$\frac{dm_f}{dt} = -\frac{dm_d}{dt} - A_m \rho J_w = -\dot{V}_o - A_m \rho J_w \quad (5)$$

Afterward, the diafiltrate continues to enter the stirred cell at a slower, steady rate for the remainder of the experiment.

Ideally, following the startup, the diafiltrate entering the stirred cell is equal to the permeate leaving the stirred cell. Albeit, if the connection between the head and the body of the stirred cell is not airtight, air may slowly leak out. If this occurs, diafiltrate enters the stirred cell and the solution volume in the stirred cell gradually increases in response to the changing air volume. We assume that the mass of the feed solution, m_f , changes at a constant rate, $\dot{V}$, throughout the experiment. $\dot{V}$ can be estimated using the initial and final values of m_f in eq 6. Therefore, for $t \geq t_L$ or $t \geq t_F$:

$$\frac{dm_f}{dt} = -\frac{dm_d}{dt} - A_m \rho J_w = \dot{V} \quad (6)$$

which implies eq 7:

$$\frac{dm_d}{dt} = -\dot{V} - A_m \rho J_w \quad (7)$$

The general solute balance in the stirred cell is given by eq 8:

$$\frac{d(c_f m_f)}{dt} = -c_d \frac{dm_d}{dt} - A_m \rho J_s \quad (8)$$

which leads to the following expression for the change in feed concentration, c_f in eq 9:

$$\frac{dc_f}{dt} = \frac{1}{m_f} \left(-c_d \frac{dm_d}{dt} - A_m \rho J_s - c_f \frac{dm_f}{dt} \right) \quad (9)$$

Time Correction for Permeate. Our model of the experimental system can also be refined on the permeate side of the membrane. As illustrated in Figure 1, the holdup volume represents the volume under the membrane where the permeate accumulates. This volume is modeled as a perfectly mixed reservoir.⁵⁵ The tube connecting the stirred cell to the scintillation vial is treated as a plug flow element, where the salt solution is well-mixed in the radial direction. As shown in Figure 3, after time t_0 , the holdup volume is filled first at t_H , followed by the tube filling until t_V . The time t_V marks the

moment when the first drop of permeate appears in the scintillation vial, as recorded by the balance, which measures the time-dependent weight of the vial. The time t_H is estimated by extrapolating the permeate mass data based on the time required to fill the tube (i.e., $v_{tb} \sim 0.8$ mL for a connecting tube with an inner diameter of 2.2 mm, and length of 200 mm). There is no strict sequence between t_H and t_L or t_F ; these times are determined based on experimental observations and incorporated into the models accordingly.

In our model, for $t_0 \leq t < t_H$, the mass (m_h) and concentration of the solution in the holdup region are given by eqs 10 and 11:

$$\frac{dm_h}{dt} = A_m \rho J_w \quad (10)$$

$$\begin{aligned} \frac{d(c_h m_h)}{dt} &= A_m \rho J_s \\ \Rightarrow \frac{dc_h}{dt} &= \frac{A_m \rho}{m_h} (J_s - c_h J_w) \end{aligned} \quad (11)$$

For $t_H \leq t < t_V$, the mass the solution in the holdup region no longer changes, eq 12 and hence the concentration is derived as eq 13:

$$\frac{dm_h}{dt} = 0 \quad (12)$$

$$\begin{aligned} \frac{d(c_h m_h)}{dt} &= A_m \rho J_s - c_h A_m \rho J_w \\ \Rightarrow \frac{dc_h}{dt} &= \frac{A_m \rho}{m_h} (J_s - c_h J_w) \end{aligned} \quad (13)$$

In the model, we neglect the time delay associated with transferring the permeate from the holdup volume to the scintillation vial and assume that the permeate within the scintillation vial is well-mixed. Therefore, for $t \geq t_V$, we have eqs 14–17:

$$\frac{dm_h}{dt} = 0 \quad (14)$$

$$\frac{dc_h}{dt} = \frac{A_m \rho}{m_h} (J_s - c_h J_w) \quad (15)$$

$$\frac{dm_v}{dt} = A_m \rho J_w \quad (16)$$

$$\begin{aligned} \frac{d(c_v m_v)}{dt} &= \frac{dm_v}{dt} c_h = A_m \rho J_w c_h \\ \Rightarrow \frac{dc_v}{dt} &= -A_m \rho J_w \frac{(c_v - c_h)}{m_v} \end{aligned} \quad (17)$$

where the efflux from the holdup volume is collected directly into the scintillation vial, eq 17.

The average concentrations of permeate collected in the scintillation vials are measured using inductively coupled plasma-optical emission spectroscopy (ICP-OES) after the experiment. To account for the time delay introduced by the connecting tube, the measurement time for the permeate concentration is corrected by backtracking the time-series vial mass data to account for the permeate mass in the tube. For example, instead of using the time stamp at the end of each vial collection, we use the time corresponding to when the total vial

mass minus the tube mass m_{tb} was reached. These time corrections ensure a direct comparison between the model-predicted vial concentrations and the experimental measurements.

Concentration Polarization. Accounting for concentration polarization on the feed side of the membrane is essential for accurately determining membrane transport coefficients. Namely, the solute concentration difference across the membrane is represented by $\Delta c = c_{in,f} - c_h$, where $c_{in,f}$ is the feed-side concentration at the solution-membrane interface, and c_h is the concentration in the holdup region, approximating the permeate concentration at the membrane-solution interface. $c_{in,f}$ accounts for the effect of concentration polarization, and is related to bulk feed/retentate concentration c_f using the thin-film model as illustrated in eqs 18 and 19.⁵⁹

$$\frac{c_{in,f} - c_h}{c_f - c_h} = \exp\left(\frac{J_w}{k}\right) \Leftrightarrow c_{in,f} = \left(c_f - c_h\right) \exp\left(\frac{J_w}{k}\right) + c_h \quad (18)$$

$$\frac{kb}{D_s} = 0.23 \left(\frac{bv^0}{\nu}\right)^{0.57} \left(\frac{\nu}{D_s}\right)^{0.33} \Leftrightarrow k = 0.23 \frac{(\nu^0)^{0.57} D_s^{0.67}}{\nu^{0.24} b^{0.43}} \quad (19)$$

where k is the mass transfer coefficient, b is the stirred cell diameter, D_s is the diffusion coefficient of the solute in water, ν is the kinematic viscosity of water, and ν^0 is the average velocity within the system, defined as the product of the stir bar angular velocity and radius.

Membrane Transport Models. The transport properties of the membrane are characterized by modeling the solvent and solute fluxes through the membrane. Specifically, the volumetric flux of water J_w is described as a function of the applied pressure difference across the membrane, eq 20⁵⁷:

$$\begin{aligned} J_w &= L_p (\Delta P - \sigma \Delta \pi) \\ &= L_p (\Delta P - \sigma n R T \Delta c) \end{aligned} \quad (20)$$

where L_p is the hydraulic permeability, a key parameter quantifying the membrane's ability to transport water, and σ is the reflection coefficient, a parameter related to the membrane's solute rejection properties. ΔP is the applied pressure in the diafiltration experiment, and $\Delta \pi = n R T \Delta c$ is the osmotic pressure, calculated using the van't Hoff equation, where n is the number of dissolved species formed by the solute(s), R is the gas constant, and T is the temperature, and Δc is the solute concentration difference across the membrane.

When ion transport across the membrane is governed by the combined effects of convection and diffusion, the local solute flux in the membrane, j_s , is mathematically described as eq 21:

$$j_s = -D_m \frac{dc(z)}{dz} + J_w c(z) \quad (21)$$

where the first term, $-D_m \frac{dc(z)}{dz}$, represents the diffusive contribution to the flux, which is driven by the concentration gradient, $\frac{dc(z)}{dz}$, with D_m being the solute diffusion coefficient in the membrane phase. The second term, $J_w c(z)$, corresponds to the convective contribution to the flux, driven by the volumetric solvent flux, J_w , which carries the solute. Because the membrane responds more rapidly than the rate at which the feed concentration changes, a pseudosteady-state approximation

was applied. This assumption yields the following differential equation for the solute concentration profile across the membrane, eq 22:

$$\frac{dc(z)}{dz} = 0, \text{ so } \frac{dj_s}{dz} = 0 \Rightarrow -D_m \frac{d^2c(z)}{dz^2} + J_w \frac{dc(z)}{dz} = 0 \quad (22)$$

which can be solved given boundary conditions on both sides of the membrane with thickness L , eqs 23 and 24:

$$c(0) = c_{in,f}^m = H_f \cdot c_{in,f} \quad (23)$$

$$c(L) = c_{in,p}^m = H_p \cdot c_{in,p} \quad (24)$$

where $c_{in,f}^m$ and $c_{in,p}^m$ are the boundary concentrations in the membrane phase on the feed and permeate sides, respectively, and are related to the interface concentrations $c_{in,f}$ and $c_{in,p}$ by the partition coefficients H_f and H_p , which account for the solute partitioning at membrane surfaces on the feed and permeate sides, respectively.

By solving for the concentration profile $c(z)$ and applying eq 21, an expression for the total molar flux of solute in terms of the Peclet number, $Pe = \frac{J_w L}{D_m}$, is developed:

$$J_s = J_w \cdot \left(\frac{c_{in,f} \cdot H_f \cdot \exp\left(\frac{J_w L}{D_m}\right) - H_p \cdot c_{in,p}}{\exp\left(\frac{J_w L}{D_m}\right) - 1} \right) \quad (25)$$

In diffusion-dominated regimes, $\frac{D_m}{L} \gg J_w$, convection becomes negligible compared to diffusion, and $\frac{J_w L}{D_m} \rightarrow 0$. Using a Laurent series expansion in this limit, J_s in eq 25 simplifies to eq 26:

$$J_s = J_w \cdot \left(\frac{H_f \cdot c_{in,f} - H_p \cdot c_{in,p}}{\frac{J_w L}{D_m}} + \frac{H_f \cdot c_{in,f} + H_p \cdot c_{in,p}}{2} + \frac{\frac{J_w L}{D_m} (H_f \cdot c_{in,f} - H_p \cdot c_{in,p})}{12} + O\left(\left(\frac{J_w L}{D_m}\right)^3\right) \right) \quad (26)$$

Approximating J_s with the first term in the series, and assuming $H_f = H_p = H$, we have eq 27:

$$J_s \approx J_w \cdot H \cdot \left(\frac{c_{in,f} - c_{in,p}}{\frac{J_w L}{D_m}} \right) = \frac{D_m H}{L} (c_{in,f} - c_{in,p}) \quad (27)$$

By defining a solute permeability coefficient, $B = \frac{D_m H}{L}$, the solute flux J_s is expressed as eq 28:

$$J_s = B \Delta c \quad (28)$$

This linear relationship highlights how the solute flux depends on the concentration gradient across the membrane when diffusion dominates. For instance, previously studied NF90 membranes exhibit transport properties that can be accurately described by this form of the model.

In regimes where diffusion and convection are both relevant mechanisms, $\frac{J_w L}{D_m} \sim 1$. Here, J_s in eq 25 can be approximate with Taylor series expansion, eq 29:

$$\begin{aligned} J_s &= J_w \cdot \left[H_f \cdot c_{in,f} + \frac{1}{e-1} (H_f \cdot c_{in,f} - H_p \cdot c_{in,p}) \right. \\ &\quad - \frac{e}{(e-1)^2} \left(\frac{J_w L}{D_m} - 1 \right) (H_f \cdot c_{in,f} - H_p \cdot c_{in,p}) \\ &\quad + \frac{e(1+e)}{2(e-1)^3} \left(\frac{J_w L}{D_m} - 1 \right)^2 (H_f \cdot c_{in,f} - H_p \cdot c_{in,p}) \\ &\quad \left. + O\left(\left(\frac{J_w L}{D_m} - 1\right)^3\right) (H_f \cdot c_{in,f} - H_p \cdot c_{in,p}) \right] \\ &= J_w \left[\frac{H_f \cdot c_{in,f}}{H_f \cdot c_{in,f} - H_p \cdot c_{in,p}} + \frac{1}{e-1} - \frac{e}{(e-1)^2} \right. \\ &\quad \left(\frac{J_w L}{D_m} - 1 \right) \\ &\quad + \frac{e(1+e)}{2(e-1)^3} \left(\frac{J_w L}{D_m} - 1 \right)^2 + O\left(\left(\frac{J_w L}{D_m} - 1\right)^3\right) \right] \Delta c \quad (29) \end{aligned}$$

Noticing the term representing the concentration difference within the membrane $\Delta c = H_f \cdot c_{in,f} - H_p \cdot c_{in,p}$ in the series, this expression reveals an apparent solute permeability coefficient that becomes dependent on feed-side concentration $c_{in,f}$ and water flux J_w :

$$\begin{aligned} J_s &= B(J_w, H_f, H_p, c_{in,f}, c_{in,p}) \cdot \Delta c, \\ B(J_w, H_f, H_p, c_{in,f}, c_{in,p}) &= J_w \left[\frac{H_f c_{in,f}}{H_f c_{in,f} - H_p c_{in,p}} + \frac{1}{e-1} \right. \\ &\quad - \frac{e}{(e-1)^2} \left(\frac{J_w L}{D_m} - 1 \right) \\ &\quad + \frac{e(1+e)}{2(e-1)^3} \left(\frac{J_w L}{D_m} - 1 \right)^2 \\ &\quad \left. + O\left(\left(\frac{J_w L}{D_m} - 1\right)^3\right) \right] \quad (30) \end{aligned}$$

Most studies assume $H \approx H_f \approx H_p$, which allows for a single H to be extracted from eq 30. As shown in the results, we found maintaining distinct partition coefficients better explains the experimental data. Next, we approximate the dependency of B on concentration as

$$B(J_w, H_f, H_p, c_{in,f}, c_{in,p}) \approx B(J_w, c_{in,f}) = J_w \sum_{i \in I} \beta_i^i c_{in,f}^i \quad (31)$$

where different polynomial orders, $I \in [\{0\}, \{0, -0.5\}, \{0, 0.5\}, \{0, 1\}, \{0, 1, 2\}, \{0, 1, 2, 3\}]$, are considered for eq 31, capturing variations in permeability due to convective flux and solute-membrane interactions in a multiplicative form. This extended model $B(J_w, c_{in,f})$ acknowledges that solute permeability is influenced by feed-side concentration, particularly in surface-charged membranes, where electrostatic interactions alter solute partitioning and transport rates.

Finally, the governing flux equations, coupled with the ordinary differential equations (ODEs) that capture states in

the diafiltration experiment, form a system of differential-algebraic equations (DAEs). This DAE model facilitates the simulation and prediction of diafiltration dynamics, enabling us to comprehensively characterize membrane behavior under varying conditions, bridging fundamental transport mechanisms with experimental observations.

MATERIALS AND METHODS

Materials. All experiments were conducted using NF270 membranes supplied by DuPont. Every experiment was conducted using a 10 mL Amicon stirred cell (Amicon, Burlington, Massachusetts). An Omega PX409-100GUSBH pressure transducer was used to monitor the pressure. Conductivity data was acquired using an LFS1107 conductivity sensor (Innovative Sensor Technology, Las Vegas, Nevada) with one of two conductivity measurement boards, CN0359 (Analog Devices Incorporated, Thief River Falls, Minnesota) or EC EvalBoard LFS (Innovative Sensor Technology, Las Vegas, Nevada). Mass data was recorded by an OHAUS Adventure Series Balance (OHAUS, Parsippany, New Jersey). Pressure, conductivity, and mass data were recorded using an in-house MATLAB code. Feed and diafiltrate solutions were prepared using deionized water supplied by a Millipore water purification system (Milli-Q Advantage A10, Milli-Q, MA). Potassium chloride (KCl) along with nitric acid (HNO_3) were purchased from Sigma-Aldrich (Aldrich, St. Louis, Missouri) with purities greater than 99% and 70%, respectively.

Diafiltration Experiments with Lag and Overflow Startups. Table 1 lists the manipulated and measured variables in a typical diafiltration experiment.

Diafiltration experiments can be conducted in two startup modes, lag or overflow, by controlling the timing at which the diafiltrate enters the stirred cell. To achieve this control, an initial pressure to the diafiltrate tank and the head space of the stirred cell is set at approximately 1/6 of the operating pressure (e.g., 10 psi for an operating pressure of 60 psi). In lag mode, the initial pressure to the whole system is set slightly above this fraction at ~ 12 psi. The valve to the stirred cell is then closed and the diafiltrate tank is pressurized to the operating pressure. Due to the compression of gas molecules within the stirred cell head space, the operating pressure is not sufficient to bring the diafiltrate to the entrance of the stirred cell. Consequently, this condition allows the solution to permeate before diafiltrate enters the stirred cell, effectively resembling a filtration experiment. During this phase, the concentration within the stirred cell increases because of the finite rejection of the solute. On the other hand, in overflow mode, the initial pressure in the headspace is set below 1/6 of the operating pressure at ~ 8 psi. Once the main valve to the stirred cell is closed, pressure is applied to the diafiltrate tank and a bolus of diafiltrate enters the stirred cell abruptly, which is detected as a spike in conductivity. This spike is caused by a sudden change in concentration.

Inline Conductivity Probe Calibration. The LFS1107 conductivity sensor was calibrated using 14 KCl stock solutions between 0 and 100 mM. Once submerged in solution, the probe was given 10 s to stabilize, and the conductivity reading given on the CN0359 conductivity board/EC EvalBoard LFS conductivity board was recorded. The probe was thoroughly rinsed with deionized water between each stock solution measurement. Linear regression was used to obtain a calibration curve (SI Figure S1) with R^2 values greater than 0.99. Using this calibration curve, the conductivity readings over the course of

Table 1. Manipulated and Measured Variables in the Diafiltration Experiments

description	variable	value
manipulated variables		
initial feed concentration	$c_f(t_0)$	e.g., 15 mM
diafiltrate concentration	c_d	e.g., 120 mM
initial applied pressure	ΔP_i	e.g., 8 psi
applied operating pressure	ΔP_t	e.g., 60 psi
operating temperature	T	25 °C
average velocity within the stirred cell	v^0	41.89 $\text{cm}\cdot\text{s}^{-1}$
measured variables		
mass of the solution in the scintillation vials	m_v	
concentration of the solution in the scintillation vials	c_v	
concentration of the feed solution	c_f	
initial mass of the feed solution in the stirred cell	$m_f(t_0)$	e.g., 10 g
final mass of the solution in the stirred cell	$m_f(t_{\text{end}})$	e.g., 10 g
known constants		
membrane area	A_m	4.1 cm^2
number of dissolved species for KCl	n	2
diffusion coefficient of K^+ in water	D_s	$1.960 \times 10^{-5} \text{ cm}^2\cdot\text{s}^{-1}$
kinematic viscosity of water	ν	$8.927 \times 10^{-3} \text{ cm}^2\cdot\text{s}^{-1}$
density of the water	ρ	1 $\text{g}\cdot\text{cm}^{-3}$
mass of solution in holdup	m_h	0.25 g
volume of solution in the tube between the stirred cell and scintillation vial	v_{tb}	0.8 mL
volume of air between the diafiltrate tank and the stirred cell	$v_{\text{tb,d}}$	35 mL
volume of the air in the headspace of the stirred cell	v_{head}	7 mL

the experiment can be related to the bulk concentration within the stirred cell.

ICP-OES Sample Preparation. A total of 14 solutions were collected in scintillation vials by the end of the experiment, including 10 permeate vials, the initial feed, final retentate, final tube, and diafiltrate solutions. These samples were analyzed using ICP-OES (PerkinElmer Avio 200). The solutions were diluted so that the fixed concentration analyzed for ICP was approximately 10 ppm. For example, higher concentration samples such as the retentate and diafiltrate were prepared by diluting 0.012 mL of solution in 5 mL of 3% nitric acid. 0.12 mL of the initial feed solution and 0.025 mL of the permeate solutions were diluted in 5 mL of 3% nitric acid, according to their respective concentrations.

Nonlinear Parameter Estimation and Uncertainty Quantification. We now describe the model-based data analytics workflow, which is summarized in Figure 4.

Parameter estimation for the DAE models was performed using a weighted least-squares (WLS) approach, which accounts for heteroscedasticity in different types of data.⁶⁰ This method ensures that data points with higher measurement precision contribute more to the parameter estimation than those with larger uncertainties.

Let $\hat{y}_r(t) = f(\mathbf{x}(t), \mathbf{u}(t), \boldsymbol{\theta})$ represent the model prediction for experimental observation $y_r(t)$ at time t , where $\mathbf{x}(t)$ assembles state variables, $\mathbf{u}(t)$ are controlled experimental conditions, $\boldsymbol{\theta}$ include all model parameters, and $r \in R$ indexes the type of response. The model assumes that measurement errors are independent and normally distributed but may exhibit non-constant variance (heteroscedasticity) across different types of

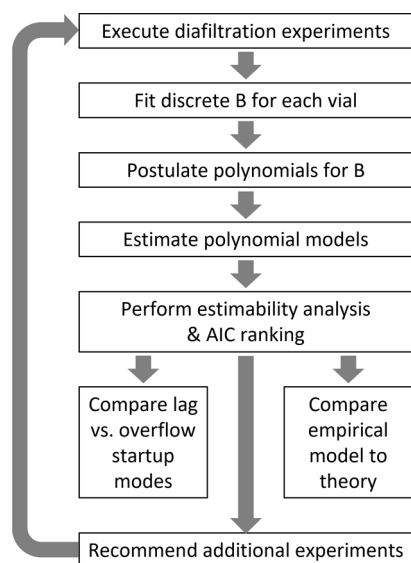


Figure 4. Model-based data analytics workflow for characterizing transport properties of surface-charged membranes.

measurements. Specifically, the error, $\epsilon_{r,i}$, associated with i -th observation, $y_{r,i}(t_i)$, is described as eq 32:

$$y_{r,i}(t_i) = \underbrace{f(\mathbf{x}(t_i), \mathbf{u}(t_i), \boldsymbol{\theta}) + \epsilon_{r,i}}_{\hat{y}_{r,i}(t_i)} \quad \epsilon_{r,i} \sim \mathcal{N}(0, s_{r,i}^2) \quad (32)$$

where $\mathcal{N}(0, s_{r,i}^2)$ denotes a normal distribution with mean zero and variance $s_{r,i}^2$. The nonconstant variance is incorporated into the model by scaling the data or applying weights in the WLS method. In this study, the variance of mass data is assumed constant, based on instrument precision, $(0.01 \text{ g})^2$, while the variance for concentration measurements is estimated from the instrument calibration curve. For instance, we assume that the standard deviation of the concentration measurement error is proportional to the measurement magnitude, eq 33:

$$s_{r,i}^2 = \alpha^2 y_{r,i}^2 \quad (33)$$

where α is a proportionality constant, 3% for ICP-OES concentration measurements, and 0.3% for concentration measurements via conductivity probe.

The scaled residuals $z_{r,i}$ are computed by dividing the residuals by their corresponding assumed standard deviations $\tilde{s}_{r,i}$, eq 34:

$$z_{r,i} = \frac{y_{r,i} - \hat{y}_{r,i}}{\tilde{s}_{r,i}} \quad (34)$$

The WLS objective function for nonlinear parameter estimation is then formulated as the sum of squared scaled residuals, normalized by the number of observations n_r for each response type r , eq 35:

$$\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta}_{\text{lb}} \leq \boldsymbol{\theta} \leq \boldsymbol{\theta}_{\text{ub}}} \sum_{r=1}^R \frac{1}{n_r} \sum_{i=1}^{n_r} z_{r,i}^2 \quad (35)$$

In this formulation, membrane property parameters are constrained within specific bounds as $\boldsymbol{\theta}_{\text{lb}}$ and $\boldsymbol{\theta}_{\text{ub}}$: $L_p \in [0.5, 50] \text{ L}\cdot\text{m}^{-2}\cdot\text{h}^{-1}\cdot\text{bar}^{-1}$, $B \in [0, 30] \mu\text{m}\cdot\text{s}^{-1}$, and $\sigma \in [0, 1]$. Minimizing objective function eq 35 allows for the estimation of

parameters while accounting for varying levels of uncertainty and data abundance across experimental observations.

The uncertainty in the parameter estimates $\hat{\boldsymbol{\theta}}$ is quantified via the covariance matrix $\mathbf{V}_{\hat{\boldsymbol{\theta}}}$,⁶⁰ eq 36:

$$\mathbf{V}_{\hat{\boldsymbol{\theta}}} \approx (\mathbf{Q}_y^T \boldsymbol{\Sigma}_y^{-1} \mathbf{Q}_y)^{-1} = (\mathbf{Q}_z^T \boldsymbol{\Sigma}_z^{-1} \mathbf{Q}_z)^{-1} \quad (36)$$

where $\mathbf{Q}_y$ is the matrix of local sensitivities of the model predictions, i.e., partial derivatives of the model predictions with respect to each parameter⁶¹ evaluated at the parameter estimate $\hat{\boldsymbol{\theta}}$. $\boldsymbol{\Sigma}_y$ is the covariance matrix of measurement errors, which is diagonal under the independence assumption. The covariance matrix can also be approximated by the sensitivity matrix of scaled residuals with respect to the parameters $\mathbf{Q}_z$.

Fisher Information Criteria for Parameter Precision.

We employed Fisher Information Matrix (FIM) analysis to determine the most informative experimental conditions to improve parameter estimation. The FIM, denoted as $\mathbf{M}$, quantifies the amount of information that an experiment provides about the model parameters. For a model with parameter vector $\boldsymbol{\theta}$, the FIM can be approximated as the inverse of the covariance matrix,⁴⁵ eq 37:

$$\mathbf{M} \approx \mathbf{V}_{\hat{\boldsymbol{\theta}}}^{-1} \quad (37)$$

In MBDoe, the FIM is employed to guide experimental design by selecting experimental conditions that maximize

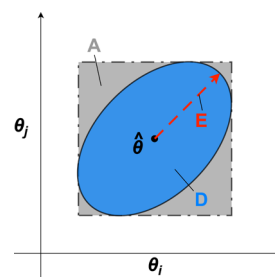


Figure 5. Geometrical interpretation of the FIM metrics with confidence ellipsoid (adapted from Franceschini and Macchietto).⁴⁵ The black dot represents parameter estimates $\hat{\boldsymbol{\theta}}$, and the blue area represents the confidence region of the parameter (in 2-D). A-optimality evaluates the enclosing hyperbox of the confidence ellipsoid, the gray dot-dashed area in 2-D. D-optimality evaluates the ellipsoid hypervolume. E-optimality evaluates the major axis of the ellipsoid, represented in a red dashed arrow.

specific metrics of $\mathbf{M}$. Figure 5 illustrates the three most common optimality criteria based on FIM⁴⁵:

- **A-optimal** designs aim to maximize the trace of the FIM, $\max \text{tr}(\mathbf{M})$, which approximately minimizes the average variance of the parameter estimates. This criterion is useful when the goal is to reduce the overall uncertainty across all parameters.
- **D-optimal** designs seek to maximize the determinant of the FIM, $\max \det(\mathbf{M})$, which corresponds to minimizing the volume (or hypervolume for higher dimensions in $\boldsymbol{\theta}$) of the parameter confidence region. It ensures that the parameter estimates are as precise as possible by maximizing the overall information content of the experiment.
- **E-optimal** designs maximize the smallest eigenvalue of the FIM, $\max \lambda_{\min}(\mathbf{M})$, ensuring that the worst-case parameter estimate (corresponding to the longest axis in

Table 2. Parameter Results Obtained by Fitting the Empirical Model $B = J_w(\beta_0 + \beta_1 c_{in,r})^a$

mode	parameters	units	estimates	std. errors	correlation matrix
lag	L_p	$\text{L}\cdot\text{m}^{-2}\cdot\text{h}^{-1}\cdot\text{bar}^{-1}$	11.11	6.78×10^{-2}	$\begin{bmatrix} 1.00 & -0.91 & 0.88 & 0.95 & -0.90 \\ -0.91 & 1.00 & -0.97 & -0.83 & 0.74 \\ 0.88 & -0.97 & 1.00 & 0.88 & -0.65 \\ 0.95 & -0.83 & 0.88 & 1.00 & -0.77 \\ -0.90 & 0.74 & -0.65 & -0.77 & 1.00 \end{bmatrix}$
	β_0	dimensionless	1.06	5.92×10^{-2}	
	β_1	mM^{-1}	0.0152	6.48×10^{-4}	
	σ	dimensionless	1.00	1.25×10^{-2}	
	$\dot{V}$	$\text{g}\cdot\text{h}^{-1}$	0.17	4.67×10^{-2}	
overflow	L_p	$\text{L}\cdot\text{m}^{-2}\cdot\text{h}^{-1}\cdot\text{bar}^{-1}$	11.57	3.11×10^{-1}	$\begin{bmatrix} 1.00 & -0.99 & 0.99 & 1.00 & 0.32 & -0.95 \\ -0.99 & 1.00 & -1.00 & -0.98 & -0.30 & 0.94 \\ 0.99 & -1.00 & 1.00 & 0.99 & 0.33 & -0.91 \\ 1.00 & -0.98 & 0.99 & 1.00 & 0.36 & -0.93 \\ 0.32 & -0.30 & 0.33 & 0.36 & 1.00 & -0.13 \\ -0.95 & 0.94 & -0.91 & -0.93 & -0.13 & 1.00 \end{bmatrix}$
	β_0	dimensionless	0.34	8.36×10^{-2}	
	β_1	mM^{-1}	0.0121	8.80×10^{-4}	
	σ	dimensionless	1.00	3.03×10^{-2}	
	$\dot{V}_o$	$\text{g}\cdot\text{s}^{-1}$	-0.16	1.71×10^{-4}	
	$\dot{V}$	$\text{g}\cdot\text{h}^{-1}$	-0.99	8.83×10^{-2}	

^aThe parameters are regressed using a lag and an overflow experiment, respectively. Data are provided in Figure 6.

confidence ellipsoid) has as much precision as possible. This criterion is particularly useful for preventing situations where one parameter is estimated with significantly less precision than others.

The selection of an optimality criterion depends on the specific goals of the experiment. In this work, we evaluate all three optimality criteria to determine better experimental operations that maximize the information content and improve the precision of parameter estimates.

Akaike Information Criterion for Model Selection. To compare different candidate models and select the one that best describes the experimental data, we employed the Akaike Information Criterion (AIC).⁶² The AIC provides a robust method for model selection by balancing model fit with model complexity, thus preventing overfitting. Specifically, it penalizes models that include unnecessary parameters, ensuring the chosen model is both parsimonious and accurate.

The AIC is computed using the likelihood function of the model, $\mathcal{L}(\theta|\mathbf{y})$, which denotes the probability of observing the data $\mathbf{y}$ given the model parameters θ , eq 38:

$$\text{AIC} = -2\ln(\mathcal{L}(\hat{\theta}|\mathbf{y})) + 2k \quad (38)$$

where k is a measure of model complexity. The first term rewards models that fit the data well, while the second term penalizes more complex models with a greater number of parameters. AIC provides a measure of the relative quality of different candidate models (i.e., only differences in AIC are meaningful for model comparisons) for the same data set and is widely used for guiding model selection. Using scaled residuals $z_{r,i}$ and assuming each type of measurement has an independent and identical error structure for R types of measurements, the AIC can be reformulated in a more convenient form ΔAIC in eq 39 by dropping the constant term:

$$\Delta\text{AIC} = \sum_{r=1}^R \left(n_r \ln \left(\frac{1}{n_r} \sum_{i=1}^{n_r} z_{r,i}^2 \right) \right) + 2k, \quad k = p + R \quad (39)$$

where the model complexity, k , is the sum of the number of model parameters, p , and the number of error structure types assumed (associated with types of measurements), R . This form is equivalent to evaluating ΔAIC for original residual $y_{r,i} - \hat{y}_{r,i}$ (see SI for derivations). The ΔAIC formula shows that a model with lower residual squares will have a lower AIC value, but adding more parameters (increasing k) will increase the AIC unless the additional parameters significantly improve the model fit.

We employed Akaike weights, which quantify the likelihood of a particular model being the best among a set of candidate models,^{63,64} to evaluate model probabilities. The Akaike weight for model j is given by eq 40:

$$w_j = \frac{\exp(-0.5\Delta_j)}{\sum_{c=1}^C \exp(-0.5\Delta_c)},$$

$$\Delta_j = \text{AIC}_j - \text{AIC}_{\min}$$

$$= \Delta\text{AIC}_j - \Delta\text{AIC}_{\min} \quad (40)$$

where Δ_j is the AIC difference between model j and the minimum calculated AIC value, $\text{AIC}_{\min}$ (or ΔAIC difference). When $\Delta < 2$, the models are considered statistically indistinguishable while $\Delta > 10$ recommends that the model with the lower AIC is strongly preferred.⁶⁵ Thus, maximizing Akaike weights serves as a criterion for discriminating between models and guiding experimental design.

In this work, AIC was used to compare different models for the apparent solute permeability coefficient B (as defined in eq 31). The model with the lowest ΔAIC was selected as the most parsimonious model for describing the experimental data, balancing the goodness-of-fit with model complexity.

Computational Environment. The parameter estimation and model simulations were implemented using the open-source Pyomo package (version 6.8.0)⁶⁷ in Python (version 3.11.6). The DAEs describing the diafiltration experiment — eqs 2 and 3 (lag mode) or eqs 4 and 5 (overflow mode), eqs 10–20, and eq 31 — were numerically integrated using the backward finite difference method provided by Pyomo.DAE.⁶⁸ These open-source packages enable transferable and reproducible model implementation, while also providing access to advanced data analysis tools and state-of-the-art optimization algorithms. For optimization, we utilized the Ipopt solver (version 3.13.2),⁶⁹ paired with the HSL MA97 linear solver,⁷⁰ both distributed via the IDAES-PSE package.⁷¹ This combination of tools allowed us to efficiently solve the complex DAE system involved in the simulations. To improve initialization and avoid potential division by zero errors, $c_h \cdot m_h$ and $c_v \cdot m_v$ were selected as combined state variables. Initial conditions for the DAE model were set as follows:

- m_f : initialized based on the initial mass of the solution in the stirred cell.
- c_f : initialized from the retentate concentration measured by the conductivity probe at the start of the experiment.

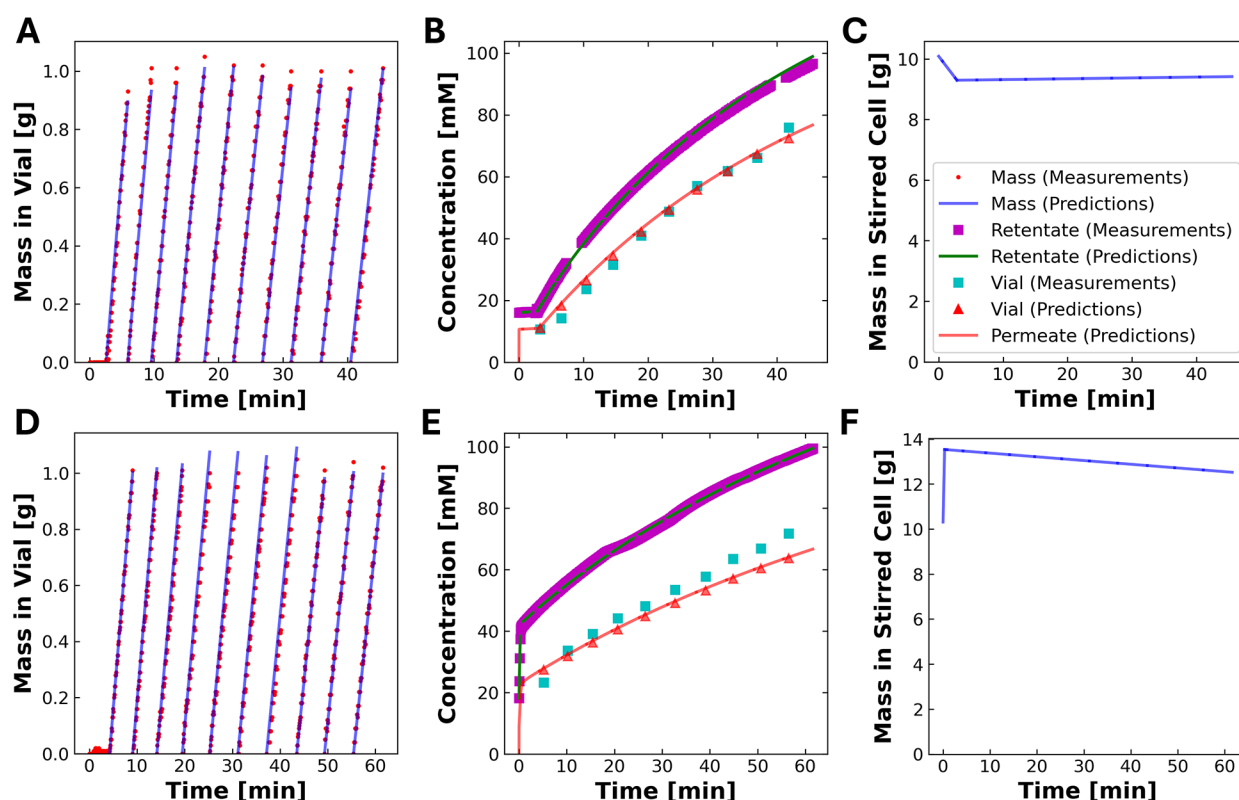


Figure 6. Plots comparing predictions (solid lines and red triangles) from diafiltration models (eqs 18–31) and experimental measurements (dots and squares). For simplicity, the error bars associated with experimental data are not included. For mass, ICP, and conductivity data, the error corresponds to the precision of the device (i.e., 0.01 g, ~3%, 0.3%, respectively). The time stamps for vial concentration measurements were corrected using the approach outlined in the section [Time Correction for Permeate](#). Diafiltration experiments were performed at 60 psi with 15 mM/120 mM KCl feed/diafiltrate solutions. Both experiments characterize the same NF270 membrane sample. (A–C) correspond to an experiment run in lag mode (refer to [A.0](#) in the SI) and (D–F) correspond to an experiment run in overflow mode (refer to [C.1](#) in the SI). Model parameters are reported in [Table 2](#).

- m_h and m_v : approximated as the mass of a single drop of solution (~ 0.05 g).
- $c_h \cdot m_h$ and $c_v \cdot m_v$: initialized to zero.

During the simulation, when vials were swapped, the model variables were reinitialized to maintain continuity. Specifically, m_b , c_b , $c_h \cdot m_h$, and $c_v \cdot m_v$ were set to the final conditions of the previous vial, while m_v was reset to the mass of one drop. A time-scaling strategy⁷² was applied to the DAEs for each vial integration, significantly improving computational efficiency by normalizing each time interval to 1. This approach dramatically reduced the computational load, achieving a simulation time of less than 1 s for a typical diafiltration experiment (compared to over 10 min without time scaling) on a standard laptop with an Intel Core i7–1065G7 CPU @ 1.30 GHz, 32 GB of RAM, and Windows 10 64-bit operating system. The combined use of advanced numerical methods, initialization strategies, and time-scaling ensured robust and efficient model performance.

RESULTS AND DISCUSSION

Empirical Correlations Effectively Model the Concentration-Dependent Permeability. Both the lag and overflow models, combined with empirical correlations for concentration-dependent solute permeability coefficient, successfully captured the transport performance of the surface-charged membranes through diafiltration experiments. The experimental conditions (i.e., diafiltrate concentration, applied pressure, and number of vial) were optimized using MBDofE to ensure the estimability of the parameters, as detailed in previous work.⁵⁶ As

will be shown later, the most appropriate polynomial model for the solute permeability coefficient is $B = J_w(\beta_0 + \beta_1 c_{in,f})$ model. For this specific model, the fitted parameters and their uncertainties are reported in [Table 2](#) for both a lag and an overflow experiment operating in the concentrating regime. Fitted parameters for all solute permeability models are included in the SI [Table S1](#). The general form of the model allows this work to be directly compared with results reported in the literature. As one example, solute permeability coefficient of KCl through NF270 membranes for feed concentrations of 67, 134, and 201 mM KCl were reported to be 20.8, 30.2, and 36.7 $\mu\text{m} \cdot \text{s}^{-1}$, respectively. While the empirically derived model in this work is distinct because it accounts for the effects of convection, the permeability coefficients regressed by the $B = J_w(\beta_0 + \beta_1 c_{in,f})$ model encompass those provided by Al-Zoubi et al.⁷³ The estimates of hydraulic permeability L_p from both experiments align well with the manufacturer's data, 13 $\text{L} \cdot \text{m}^{-2} \cdot \text{h}^{-1} \cdot \text{bar}^{-1}$ and characterization results from the literature, which report a hydraulic permeability between 8.5 and 17 $\text{L} \cdot \text{m}^{-2} \cdot \text{h}^{-1} \cdot \text{bar}^{-1}$.^{74–78}

The overflow mode, including an additional model parameter (the mass flow rate of the initial bolus of diafiltrate), $\dot{V}_o$, results in higher uncertainties in all model parameters compared to the lag mode. The correlation matrices for both modes demonstrate strong interparameter correlations, particularly between L_p , β_0 , and β_1 , which highlights the complexity of disentangling their individual contributions to the overall model performance. Notably, the higher parameter correlations obtained from the overflow experiment may originate from the loss of information

Table 3. AIC Ranking with a Diafiltration Experiment Conducted in the Concentrating Regime^a

B form	WLS objective	complexity k	Δ	Akaike weights	rank
$B = \text{const.}$	94.91	7	144.16	2.67×10^{-32}	5
$B_i = \text{const.}$	74.47	16	-117.95		
$B = J_w \sum_{i \in I} \beta_i c_{\text{in},f}^i$					
$I = \{0\}$	108.42	7	317.93	4.92×10^{-70}	7
$I = \{0, -0.5\}$	95.06	8	94.93	1.30×10^{-21}	4
$I = \{0, 0.5\}$	93.56	8	28.11	4.23×10^{-7}	3
$I = \{0, 1\}$	92.62	8	0	0.54	1
$I = \{0, 1, 2\}$	89.83	9	0.30	0.46	2
$I = \{0, 1, 2, 3\}$	83.28	10	147.41	5.25×10^{-33}	6

^a $B_i = \text{const.}$ corresponds to discrete B value for each vial collection.

at low retentate concentrations. For charged membranes, electrostatics will have the greatest effect in low ionic-strength solutions; therefore, we hypothesize that collecting information at low retentate concentrations, as is accomplished by running a lag experiment, is particularly important in characterizing membrane transport. Consequently, the lag mode, which features fewer parameters, lower parameter correlation, and reduced uncertainties, is the preferred experimental operation for membrane characterization.

Figure 6 presents the three types of measurements from diafiltration experiments. The red dots represent the mass of permeate collected in each scintillation vial (m_v), the magenta squares show the concentration of the retentate in the stirred cell (c_r), and the cyan squares depict the final concentration of permeate in each vial (c_v), with the time delay corrected for the filling of the connecting tube (as described in the Mathematical Modeling section). The solid lines in Figure 6 are continuous predictions from the calibrated models for state variables m_v (blue in Figure 6A,C), c_r (green in Figure 6B,E), c_h (red in Figure 6B,E) and m_f (blue in Figure 6C,F). Only the predictions of c_v at the end of each vial collection were plotted to facilitate comparison with post-experiment ICP-OES measurements. Figure 6A,B shows data from an experiment conducted in lag mode. Here, the model predictions closely match all three types of experimental data, particularly in capturing the time-varying concentrations. In Figure 6C, the mass of the feed solution in the stirred cell, m_b , initially decreases as no diafiltrate enters (resembling a filtration experiment). After the lag, diafiltrate begins to enter, interrupting the decrease in m_b . The later slight increase in m_b is due to air leakage. Next, for overflow mode, as shown by the concentration data in Figure 6D,E, the model overestimates the vial concentration at the beginning and underestimates it toward the end of the experiment. In Figure 6F, the rapid increase in the feed solution mass during the first few seconds reflects the rapid influx of diafiltrate. However, the ~ 2 g overflow mass prediction overshoots the actual volume of the bolus, which could be attributed to a slight delay in the conductivity probe response or incomplete solution mixing during overflow. The overflow process may have been too rapid for the well-mixed feed assumption to hold, introducing errors in measuring retentate concentration and determining the time cut for overflow (i.e., t_F). The parameter estimates for β_0 and β_1 in the overflow mode were slightly lower than those in the lag mode, whose estimates with lower uncertainties would be more reliable in describing the concentration-dependent solute permeability with higher solute flux across the membrane. In Figure 6B,E, the sharp increase in the red lines at time 0 indicates that the diafiltration model is robust, especially with the initialization of the direct permeate concentration in the holdup

region. The model performance further reinforces lag mode as a preferred startup mode for diafiltration.

SI Figures S2–S6 provide additional diafiltration experiments validating the empirical correlation for the apparent solute permeability coefficient based on lag mode results shown in Table 2. $B(J_w, c_{\text{in},f})$ aligns well with the experimental mass and concentration measurements, with bias in permeate prediction observed only at low concentrations and low pressures. This bias may be due to limitations where the diffusive rate becomes faster than the convective rate in low-pressure experiments or in the assumption of a constant reflection coefficient, σ , used in the model. In surface-charged membranes such as the NF270, the reflection coefficient may also vary with operating conditions due to changes in electrostatic interactions with ionic strength. At low concentrations, stronger electrostatic repulsion enhances solute rejection, while at higher concentrations, charge shielding reduces the influence of the membrane surface charge. Extending the model to include a concentration-dependent reflection coefficient could provide more accurate predictions, particularly for low-concentration regions, by capturing nonlinear rejection behavior and dynamic solute-membrane interactions. Incorporating more flexible membrane parameter models is an important area for future work, especially when scaling up to larger systems or working with more complex membrane structures.

Diafiltration in the Diluting Regime Improves Information Gain for Model Discrimination. Prior to fitting a concentration-dependent solute permeability model, such as eqs 30 and 31, we estimated discrete solute permeability values for each permeate vial collection during the experiments. The estimates revealed broadly spread B values ranging between 4 and $30 \mu\text{m}\cdot\text{s}^{-1}$ ($30 \mu\text{m}\cdot\text{s}^{-1}$ was set as the upper bound) across different ranges between 15 and 120 mM of feed-side solution-membrane interface concentrations, as shown in SI Figure S7A. To investigate the potential impact of water flux J_w on solute permeability, we normalized the solute fluxes J_s by J_w and plotted them against the interface concentration in SI Figure S7B. The resulting trends converged toward a common concentration-dependent relationship, indicating that solute permeability is influenced by both water flux J_w and feed-side concentration. Based on this observation and the Taylor series expansion in eq 29, we postulated concentration-dependent permeability relationships eq 31, and employed the AIC to compare alternative models.

Table 3 presents the AIC rankings for various forms of B , eq 31, fitted to data from a lag mode diafiltration experiment conducted in the concentrating regime (Figure 6A,B). Although more complex models provide a better fit in terms of the WLS objective, their higher complexity leads to a penalty in the AIC ranking. The model that assigns a discrete B value to each vial

Table 4. AIC Ranking with a Diafiltration Experiment Conducted in the Diluting Regime

B form	WLS objective	complexity k	Δ	Akaike weight	rank
$B = J_w \sum_{i \in I} \beta_i c_{in,f}^i$	$I = \{0, -0.5\}$	8	1740.82	0	4
	$I = \{0, 0.5\}$	8	830.96	3.62×10^{-181}	2
	$I = \{0, 1\}$	8	0	1	1
	$I = \{0, 1, 2\}$	9	968.13	5.94×10^{-211}	3

collection, has the most parameters but suffers from estimability issues due to limited information associated with a single vial collection. Consequently, despite achieving a better fit, this model is not considered a viable option in the ranking. The first- and second-order polynomials for B rank 1 and 2, respectively, with Akaike weights of 0.54 and 0.46, and a Δ value of 0.30, indicating that these two models are statistically indistinguishable when using just the concentrating regime diafiltration experimental data. While the concentrating diafiltration experiment presented in Figure 6A,B provides sufficient information for parameter estimation, it falls short for complete model discrimination. In contrast, Table 4 reports AIC rankings from a diafiltration experiment conducted in the diluting regime, which demonstrates enhanced model discrimination capability. This experimental design significantly improves the information for model selection, as evidenced by the higher Akaike weight of the top-ranked model and larger Δ values between competing models. Specifically, in the diluting diafiltration, the model $B = J_w(\beta_0 + \beta_1 c_{in,f})$ provides the best fit, with an Akaike weight of 1, whereas the more complex model fails to offer a substantial improvement in the fit.

This improvement in model discrimination achieved by operating in the diluting regime likely stems from the greater number of data points collected at low retentate concentrations. For example, when diluting experiments are conducted with a 30 mM/1 mM KCl feed/diafiltrate solution, the retentate concentration gradually decreases from 30 to 15 mM KCl over the 40 min experiment (Figure S6, E.1). In contrast, concentrating regime experiments use a high-concentration diafiltrate, allowing the retentate concentration to increase rapidly but reducing the amount of time spent at each concentration. For instance, the lag experiment shown in Figure 6 traverses the 15–30 mM KCl range in under 10 min. Consequently, the concentrating regime produces approximately 75% fewer data points across this critical concentration range. As will be highlighted in the subsequent section, data at retentate concentrations below the charge density of the membrane may be particularly important, as electrostatic interactions dominate transport in this regime. Therefore, although diluting experiments proceed more slowly, they yield more information in the concentration range where solute permeability is more sensitive to feed concentration—enhancing their ability to discriminate between competing models.

Previous studies have mainly focused on using MBDoE to improve parameter precision in concentrating diafiltration experiments,⁵⁶ while the AIC results here emphasize the potential of diafiltration in the diluting regime—used in conjunction with MBDoE—to optimize experimental conditions for model discrimination. In cases where a larger set of candidate models is involved, MBDoE can further enhance the identification of the dominant transport mechanism by selecting operating conditions that maximize Δ or Akaike weight, particularly within the diluting regime of diafiltration.

Data-Driven Model Identifies the Interfacial and Transport Phenomena. To mechanistically interpret the concentration-dependent relationship observed in eq 31, we fit the convection-diffusion model described in eq 25. The incorporation of the convective term within the model was motivated by the large degree of swelling observed in polyamide nanofiltration membranes that are exposed to humid atmospheres (up to 40–50% in polyamide membranes fabricated using a layer-by-layer polymerization).^{79,80} This high degree of swelling and associated water uptake can result in diffusion-engendered convection (i.e., frame-of-reference corrections). The high-fidelity state estimations from the empirical model in the previous subsection provided a robust foundation for further exploration of the relevant concentration-dependent factors in the convection-diffusion model. By leveraging simulated time series data for J_w , $c_{in,f}$, $c_{in,p}$ ($\approx c_h$), and J_s from the empirical model, as illustrated in Figure 7A–C (see Figure S8 for J_s vs time), we fit the convection-diffusion model for a lag experiment, as shown in Figure 6. For this analysis, we ignore the data from the first 100 s to avoid start-up dynamics.

Specifically, motivated by the empirical model, we assume the partition coefficients H_f and H_p change linearly with the interface concentrations, eq 41:

$$H_f = h_0 + h_1 \cdot c_{in,f}, \quad H_p = h_0 + h_1 \cdot c_{in,p} \quad (41)$$

Next, we substitute these expressions into eq 25 and rearrange to obtain eq 42:

$$\frac{J_s}{J_w} = \frac{(h_0 + h_1 \cdot c_{in,f}) \cdot c_{in,f} \cdot \exp(\text{Pe}) - (h_0 + h_1 \cdot c_{in,p}) c_{in,p}}{\exp(\text{Pe}) - 1}$$

$$\frac{J_s}{J_w} = h_1 \cdot \underbrace{\left(\frac{c_{in,f}^2 \exp(\text{Pe}) - c_{in,p}^2}{\exp(\text{Pe}) - 1} \right)}_{x_1} + h_0 \cdot \underbrace{\left(\frac{c_{in,f} \exp(\text{Pe}) - c_{in,p}}{\exp(\text{Pe}) - 1} \right)}_{x_0} \quad (42)$$

If we assume a constant Peclet number for an experiment, then h_0 and h_1 can be estimated via linear regression using eq 42 with input features x_0 and x_1 and output measurements y . Figure 8 reports the regressed parameters h_0 and h_1 for Peclet numbers from 10^{-3} to 10^2 . This plot shows two clear local minima for $\text{Pe} \approx 10^{-3}$ and $\text{Pe} \geq 5$, with the latter resulting in a modestly lower mean squared error. We originally attempted to simultaneously fit h_0 , h_1 , and Pe with nonlinear regression, but found the estimates sensitive to the initial guess and parameter bounds. Reformulating this as a linear regression problem and performing a sensitivity analysis was more insightful. For example, it reveals the regression objective becomes insensitive for $\text{Pe} \geq 5$ and the coefficients converge to $h_0 = 0.595$ and $h_1 = 0.0011$ mM.

The coefficient H_i , which is expressed in terms of the parameters h_0 and h_1 , reflects how solute partitions at the membrane-solution interfaces. For surface-charged NF270 membranes, electrostatic interactions and steric hindrance are hypothesized to control this partitioning. Regarding steric hindrance, the ratio of the hydrated size of K^+ (0.66 nm ⁸¹) to the size of the void elements in the NF270 membrane ranges from 0.61 to 0.85,^{75,77,81–84} leading to calculated hindrance factors of

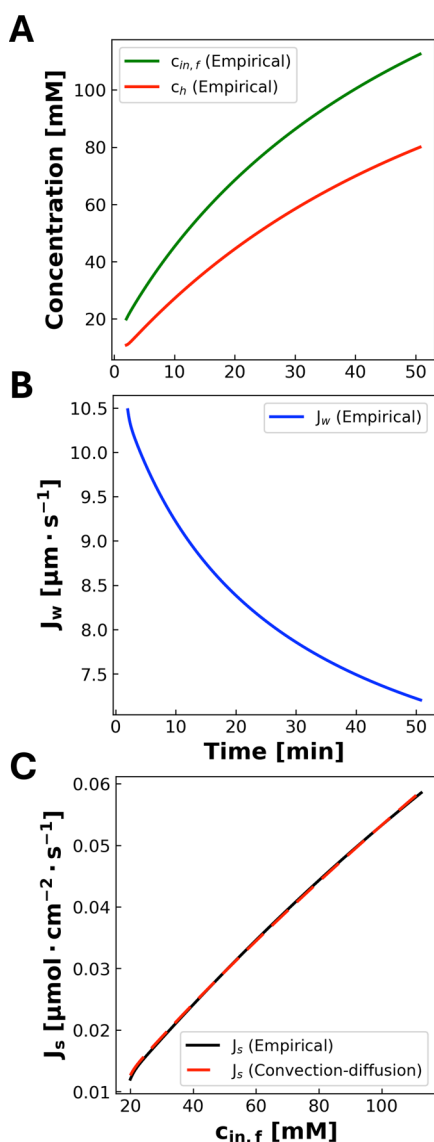


Figure 7. Plots illustrating the fitting of the convection–diffusion model (eq 25) to simulated data from the empirical model (eq 31). Solid lines in panels (A) $c_{in,f}$ and c_h , (B) J_w , and (C) J_s represent simulated data from the empirical model for a lag diafiltration experiment. The red dashed line in panel (C) depicts the solute flux J_s obtained from the convection–diffusion model fit with respect to the interface concentration. Corresponding model parameters are reported in Table 2 and Figure 8. Because the permeate holdup volume is small, we assume that c_h is approximately $c_{in,p}$.

0.044–0.28.^{77,85} These values are consistent with the order of magnitude of H established by the fitted values of h_0 and h_1 . The influence of electrostatic interactions on ion partitioning can be observed through the h_1 factor. At low concentrations, the electrostatic repulsion between the negatively charged membrane surface and the co-ion Cl^- results in reduced partitioning, yielding lower H values. As the solute concentration increases, the buildup of counterions near the membrane surface (charge shielding) mitigates these repulsive interactions, leading to higher solute partitioning at the interface. This dynamic interplay between electrostatic forces and solute concentration is effectively captured through a first-order relationship. Furthermore, this first-order relationship is consistent with the Donnan equilibrium, where the partition coefficient is described

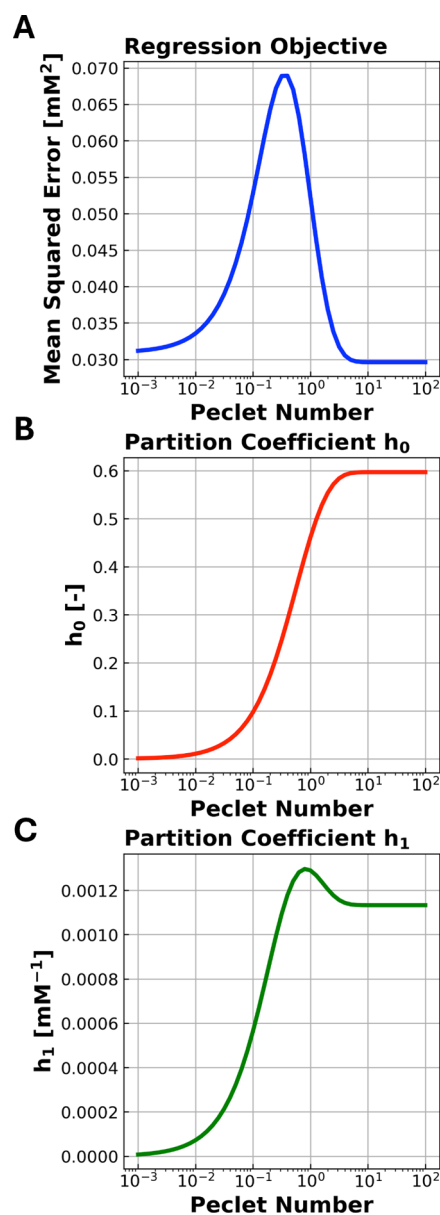


Figure 8. Mean squared error (A) and partition coefficient correlation coefficients h_0 (B) and h_1 (C) are estimated as a function of Pe . Standard errors for these regressed parameters are smaller than the line thickness. We caution against overinterpreting these standard errors, as they do not consider uncertainty in the experimental data propagated through the empirical model.

as a function of the fixed charged density χ (see derivation in SI, where χ is estimated using zeta potential measurements and the Gouy–Chapman equation⁸⁶). The analysis of the Donnan equilibrium relationship for the co-ion concentration in the membrane phase demonstrates that h_1 should be proportional to H_{steric}/χ . Using h_0 as an approximation of the steric contribution to partitioning along with a value of χ determined from zeta potential measurements, 140 mM, $h_0/\chi = 0.0043 \text{ mM}^{-1}$, which aligns well with the fitted value of h_1 . This agreement reinforces the significance of steric hindrance and electrostatic interactions in determining ion partitioning in NF270 membranes.

Next, we consider the value of the solute diffusion coefficient in the membrane inferred from the $Pe = 5$ result. J_w ranges from 7 to $10 \text{ } \mu\text{m}\cdot\text{s}^{-1}$ in the reported experiment. This implies D_m/L is between 1.4 and $2.0 \text{ } \mu\text{m}\cdot\text{s}^{-1}$. Moreover, the reported active layer

Table 5. Model Predictions Analyzed for a Lag and an Overflow Experiment^a

Mode	Model	Startup Data	Time Correction	WLS Obj.		Mass Obj.	Permeate Obj.	Retentate Obj.	
				Orig.	Trunc.			Orig.	Trunc.
Lag	M1	✓	✓	46.23	48.25	8.97	2.69	34.57	36.58
	M2	✓	✗	93.90	95.88	9.11	50.38	34.40	36.39
	M3	✗	✓	-	246.20	11.55	68.60	-	166.05
	M4	✗	✗	-	217.26	11.55	39.65	-	166.05
Overflow	M1	✓	✓	39.71	37.60	13.56	14.81	11.34	9.23
	M2	✓	✗	31.03	28.85	9.88	9.53	11.62	9.45
	M3	✗	✓	-	98.48	17.22	67.99	-	13.27
	M4	✗	✗	-	76.47	40.66	21.26	-	14.55

^aTwo important experimental factors can be incorporated in the various models: (1) the startup data and (2) the time correction for permeate vial concentrations. The four possible combinations are examined as models M1 to M4. M1 and M2 leverage the data associated with the startup processes and consider the time-varying mass in the stirred cell. Models M3 and M4 analyze truncated data sets that do not include the startup processes and assume a constant mass in the stirred cell. With regard to permeate vial concentration data, models M1 and M3 incorporate a time correction for permeate measurements. M2 and M4 do not account for this correction. All four models were used to analyze a lag and an overflow experiment.

thicknesses for NF270 membranes are between 18 and 800 nm.^{74–77,87} Thus, assuming $Pe = 5$ results in D_m estimates for KCl between 3×10^{-10} and $2 \times 10^{-8} \text{ cm}^2 \cdot \text{s}^{-1}$, which aligns with the expected range for KCl with restricted mobility in dense polymer phases.^{88–90} Delving into a more mechanistic understanding of the D_m and H estimate will require further systematic studies.

One of the key advantages of fitting the convection-diffusion model based on the empirical model simulations lies in its significant computational efficiency. Unlike the empirical model fitting in the previous subsection, which required solving a system of DAEs to account for the time-varying concentration profiles, the convection-diffusion model calibration bypasses this computational burden. By directly fitting to simulated data, it eliminates the need for explicit ODE integration, reducing parameter estimation time from over 20 min to less than 5 s. This two-step approach is particularly beneficial when screening multiple phenomenological models to mechanistically identify the phenomena that control membrane transport. It strikes a balance between physical accuracy and computational efficiency, making it a valuable tool for large-scale process design, real-time optimization, and process control in membrane-based applications.

Modeling Startups Improves Information Gain from Dynamic Diafiltration Experiments. Incorporating the two startup modes (lag and overflow) and accounting for the time-varying mass of the feed solution in the stirred cell significantly improves the performance of the diafiltration model. Table 5 compares the alignment of model predictions with experimental measurements using the WLS objectives for two experiments performed under lag and overflow modes. Four different data analysis methods were evaluated. M1 and M2 incorporate data from the process startup and use the lag or overflow model. In contrast, M3 and M4 do not model startup (i.e., they analyze truncated data collected after the first drop of permeate appeared in the scintillation vial, $t \geq t_v$) and assume a constant feed solution mass. Additionally, M1 and M3 include time correction for permeate measurements, while M2 and M4 do not. M1 provides the model predictions shown in Figure 6. In both lag and overflow modes, M1 and M2 yield smaller WLS objective values for truncated data compared to M3 and M4, as well as lower objectives for individual measurement types. For example, in the lag mode, M1 achieves a WLS objective of 48.25 for the truncated data, which is significantly lower than M3 with 246.20 and M4 with 217.26, illustrating the benefits of modeling

the startup process and accounting for the time-varying feed mass. Similarly, in the overflow mode, M1 results in a WLS objective of 37.60 for truncated data, compared to M3 with 98.48, and M4 with 76.47. This difference indicates that including the lag or overflow model enables a better fit to truncated diafiltration data while maintaining similar prediction quality for additional retentate concentration data during the startup process. For instance, in the lag mode, M1's objective for truncated data is 48.25, and the original objective value is 46.23, showing consistency in model performance across both data sets. Incorporating time correction in M1 significantly improves the permeate objective in the lag mode, reducing it from 50.38 (in M2) to 2.69. However, in the overflow mode, time correction did not produce significant improvements. This may be due to the overshooting in feed mass predictions, leading to an overestimation of permeate concentration, which subsequently aligns with the concentration in later stages.

Table 6 presents the FIM metrics, illustrating the information gain by including startup modes in the model. The A-, D-, and E-

Table 6. Information Improvements (with Respect to Parameters L_p , β_0 , β_1 , and σ) from Modeling Two Startup Modes

mode	model	metrics of FIM		
		A	D	E
		trace	determinant	minimal eigenvalue
lag	M1	2.36×10^8	5.30×10^{20}	1023.311
	improvement	8%	138%	83%
	M3	2.18×10^8	2.23×10^{20}	560.680
overflow	M1	2.78×10^9	4.11×10^{21}	164.016
	improvement	−8%	−9%	13%
	M3	3.01×10^9	4.51×10^{21}	144.867

optimality metrics (trace, determinant, and minimal eigenvalue of the FIM, respectively) quantify the information gain for parameter estimation. In the lag mode, M1, which includes startup data and time correction, shows an 8% improvement in A-optimality, a 138% improvement in D-optimality, and an 83% increase in E-optimality compared to M3, which excludes startup data. These improvements highlight the enhanced ability to infer model parameters from multiple perspectives when startup processes are modeled. In the overflow mode, however, only the E-optimality metric shows improvement, with a 13%

increase, while A- and D-optimality metrics show slight decreases when startup data are included. This suggests that the overflow model does not capture the startup process as accurately as the relative steady dynamics of the following experiment, leading to a reduction in the effective information gain.

The results in Table 6 underscore the significant role of modeling the startup processes. Lag mode experiments that model startup (i.e., M1) are the most informative. Overflow experiments or lag experiments that do not model startup processes (e.g., prior work²⁶) provide comparable amounts of information. As such, accurately capturing the early stage dynamics of lag experiments allows for more precise parameter estimation, ultimately leading to a better understanding of solute transport mechanisms. This information gain exemplifies the synergistic feedback between experimental design, model development, and data analysis, driving more informed and efficient advancements in membrane characterization.

IMPACTS AND OPPORTUNITIES

One key component toward the inverse design of materials lies in the ability to model, and subsequently identify, the mechanisms that dominate solute transport. Diafiltration experiments illustrate how researchers can identify the key factors that govern transport behavior by leveraging dynamic measurements to explore wide concentration spaces. Specifically, Figure 9 summarizes the key benefits of dynamic modeling and data analytics for this study. By capturing dynamic system behaviors and indiscriminately weighing the effects of transport phenomena (e.g., diffusion, convection, and solute partitioning), data-driven frameworks can provide a detailed understanding of how membrane structure and operating conditions affect material properties and performance. These frameworks lay the groundwork for systematically optimizing membranes for diverse applications (e.g., water desalination, biomedical separations, chemical production, wastewater treatment) where precise control over membrane properties is critical to achieve specific performance goals, providing a direct pathway to optimize selectivity, permeability, and long-term performance in complex industrial environments.

Building on the gains from diafiltration experiments and data-driven modeling, advanced experimental design and automation present opportunities to accelerate membrane innovations. Advanced optimization techniques, such as MBDoe, allow systematic exploration of membrane structure–property–performance relationships by optimizing conditions like feed concentration profiles, operating pressures, and permeate flux. Faster, targeted experiments informed by MBDoe not only reduce resource demands needed for elucidating transport mechanisms, but also enable precise fine-tuning of materials to meet specific application needs, such as ion-selective separations and molecular sieving.

Automation amplifies these capabilities by enabling high-throughput experimentation with minimal human intervention. Looking ahead, incorporating additional solute–solute and solute–membrane interactions can further expand the experimental design space. For instance, automated systems equipped with multiple diafiltrate tanks can switch between concentrating, diluting, and critical regimes.⁵⁸ In a similar manner, experiments can possess time-varying applied pressures. In this way, flexible experiments that systematically vary operating conditions can provide information-rich data sets that promote a deeper understanding of membrane performance. Real-time monitor-

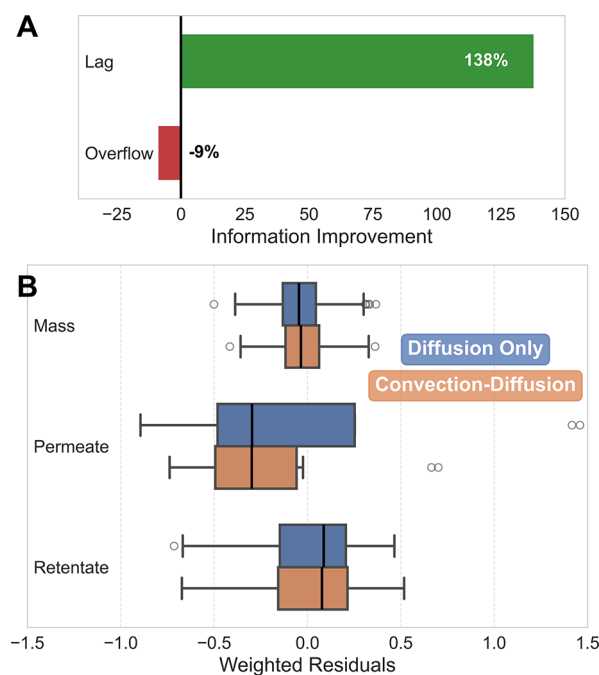


Figure 9. Modeling the dynamics of diafiltration experiments helps elucidate the relevant solute transport phenomena. (A) Lag mode diafiltration experiments that model the startup processes contain ~138% more information (quantified by D-optimality) than baseline prior work, which used lag mode experiments that do not consider startup dynamics. In comparison, overflow mode experiments are slightly less informative than this prior baseline, even with modeling startup dynamics. (B) Box plots show the distribution of weighted residuals $z_i/\sqrt{n_i}$, see eqs 34 and 35, for the three measurements in a concentrating experiment. The diffusion-only model considers a constant B , whereas the convention–diffusion model uses the best B polynomial correlation identified in this work. Accounting for convection and the concentration dependence of B enhances the accuracy of the permeate concentration predictions.

ing and in-line analytics within SDL systems further enhance experimentation by providing immediate feedback for process adjustments.

The integration of MBDoe, automation, and SDL systems in diafiltration experiments could catalyze high-throughput, data-rich investigations into membrane behavior. For instance, faster experimental cycles, informed by MBDoe, support the systematic exploration of advanced material features like dynamic covalent interactions within the polymer backbone or functional group distribution across multiple length scales.

Another key opportunity is embedding diafiltration-driven models into process-scale optimization frameworks.^{91,92} The polynomial expressions for concentration-dependence performance can act as computationally efficient reduced-order scale-bridging models^{34,93} to facilitate rigorous optimization of multiple design and operational scenarios to identify the optimal membrane system configurations. Moreover, such multiscale optimization approaches can facilitate bidirectional feedback loops across materials, devices, and process systems.¹³ For example, the proposed diafiltration experimental apparatus and modeling framework can rapidly distinguish between dominant transport phenomena and identify predictive science-based models with quantified uncertainty. These models can then be embedded in process and system optimization frameworks to assess the technical, economic, and environmental performance

of novel materials deployed at the infrastructure scale. Similarly, said process and infrastructure models can identify realistic operating conditions and quantitative materials performance targets to guide molecular engineering endeavors. These are just a few examples of how bottom-up and top-down guidance can accelerate the development of impactful membrane technologies to address challenging separations such as sustainable water treatment, industrial separations, environmental remediation, and critical mineral recycling.

CONCLUSIONS

In this work, we demonstrated the use of dynamic diafiltration experiments, combined with model-based data analytic frameworks, to characterize the transport properties of surface-charged NF membranes. By incorporating startup dynamics, time corrections, and empirical correlations between concentration and solute permeability, the proposed models provide quantitative insights into membrane behavior under various conditions. Notably, the lag startup mode outperformed the overflow mode in parameter precision, improving model accuracy and information gain by 8%, 138%, and 83% through A-, D-, and E-optimality, respectively. The improved parameter estimability stems from the ability to capture low-concentration retentate conditions, consequently modeling larger ranges of the feed concentration. Additionally, the integration of empirical and convection-diffusion models with a two-stage computational approach enabled efficient model screening without solving complex DAEs, offering mechanistic insights into concentration-dependent transport phenomena. Diafiltration experiments in the diluting regime proved effective for model discrimination, surpassing original concentrating regime diafiltration approaches used in distinguishing dominant transport mechanisms. While this study focused on posthoc model selection using AIC, future work could leverage MBDoe to proactively optimize experimental conditions for discriminating between competing transport models, improving data efficiency.

Beyond characterization, this framework supports the systematic exploration of structure–property–performance relationships, enabling the tailoring of membrane material design for specific applications, such as ion-selective separations and organic solvent filtration. Automation and real-time system monitoring into SDLs, including MBDoe approaches, could further accelerate membrane development by facilitating high-throughput experimentation and rapid experimental refinement. These advancements pave the way for robust, high-performance membranes designed for industrial separations and environmental remediation. Ultimately, this study establishes a foundational framework that combines dynamic diafiltration experiments, advanced modeling, and optimization to accelerate the development of next-generation NF membranes. These findings advance our understanding of transport mechanisms governing membrane performance while fostering innovations in material design and resource-efficient separations.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.iecr.4c04763>.

Model derivation, membrane charge density estimation, instrument calibration, and transport properties verification with additional data sets (PDF)

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Notes

The authors declare no competing financial interest.

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NOMENCLATURE

Abbreviations

AIC	Akaike information criterion
DAE	differential-algebraic equation
DATA	diafiltration apparatus for high-throughput analysis
FIM	Fisher information matrix
ICP-OES	inductively coupled plasma-optical emission spectroscopy
MBDoe	model-based design of experiments
NF	nanofiltration
ODE	ordinary differential equation
RO	reverse osmosis
SDL	self-driving laboratory

TFC	thin-film composite
UF	ultrafiltration
WLS	weighted least-squares

Data analytics

α	proportionality constant
$\hat{\theta}$	model parameter estimate
θ	model parameter vector
θ_{lb}	lower bounds for model parameters
θ_{ub}	upper bounds for model parameters
$\mathbf{M}$	Fisher information matrix
$\mathbf{Q}_y$	sensitivity matrix of model predictions
$\mathbf{Q}_z$	sensitivity matrix of scaled residuals
$\mathbf{u}$	constant/controlled experimental condition vector
$\mathbf{V}_{\theta}$	parameter covariance matrix
$\mathbf{x}$	experimental state variable vector
$\mathbf{y}$	experimental observation vector
ΔAIC	AIC dropping constant terms
Δ	AIC difference
ϵ	experimental observation error
$\hat{y}$	model prediction variable
$\lambda_{\min}(\cdot)$	smallest eigenvalue of a matrix
$\mathcal{L}(\cdot)$	likelihood function
$\mathcal{N}(\cdot)$	normal distribution
Σ_y	covariance matrix of measurement errors
Σ_z	covariance matrix of scaled measurement errors
$\text{tr}(\cdot)$	trace of a matrix
$\det(\cdot)$	determinant of a matrix
$\tilde{s}$	assumed standard deviation
$f(\cdot)$	differential or algebraic equations
k	model complexity
n_r	number of experimental observations for response r
p	number of model parameters
$p(\cdot)$	probability distribution function
R	number of response types
R^2	coefficient of determination
s	standard deviation
y	experimental observation variable
z	scaled residual

Model Variables and parameters

β	factors of polynomials in solute permeability coefficient
χ	volumetric charge density of the membrane, $\text{meq}\cdot\text{L}^{-1}$
$\Delta\pi$	osmotic pressure, bar
Δc	solute concentration difference across the membrane, $\mu\text{mol}\cdot\text{cm}^{-3}$, mM
ΔP	applied pressure, bar
ΔP_i	initial applied pressure, bar
ΔP_t	applied operating pressure, bar
$\dot{V}_o$	overflow rate of diafiltrate bolus entering the stirred cell, $\text{g}\cdot\text{s}^{-1}$
$\dot{V}$	rate of the feed solution mass change in the stirred cell, $\text{g}\cdot\text{s}^{-1}$
ν	kinematic viscosity of the solvent, $\text{cm}^2\cdot\text{s}^{-1}$
ρ	density of the solution, $\text{g}\cdot\text{cm}^{-3}$
σ	reflection coefficient, dimensionless
A_m	area of the membrane, cm^2
B	solute permeability coefficient, $\text{cm}\cdot\text{s}^{-1}$
b	diameter of the stirred cell, cm
c	solute concentration, $\mu\text{mol}\cdot\text{cm}^{-3}$, mM
c_d	concentration in the diafiltrate solution, $\mu\text{mol}\cdot\text{cm}^{-3}$, mM
c_f	concentration of the feed solution, $\mu\text{mol}\cdot\text{cm}^{-3}$, mM
c_h	concentration within hold-up volume, $\mu\text{mol}\cdot\text{cm}^{-3}$, mM

c_v	concentration of the solution in scintillation vial, $\mu\text{mol}\cdot\text{cm}^{-3}$, mM
$c_{in,f}$	concentration at the feed-side solution-membrane interface, $\mu\text{mol}\cdot\text{cm}^{-3}$, mM
$c_{in,f}^m$	concentration in the membrane phase at the feed-side boundary, $\mu\text{mol}\cdot\text{cm}^{-3}$, mM
$c_{in,p}^m$	concentration in the membrane phase at the permeate-side boundary, $\mu\text{mol}\cdot\text{cm}^{-3}$, mM
D_m	diffusion coefficient of the solute in the membrane, $\text{cm}^2\cdot\text{s}^{-1}$
D_s	diffusion coefficient of the solute in the solvent, $\text{cm}^2\cdot\text{s}^{-1}$
H	partition coefficient of the solute, dimensionless
h	factors of polynomials in partition coefficient
H_f	partition coefficient of the solute at feed side, dimensionless
H_p	partition coefficient of the solute at permeate side, dimensionless
H_{steric}	partition coefficient attribute to steric hindrance, dimensionless
I	polynomial order sets
J_s	solute flux across the membrane, $\mu\text{mol}\cdot\text{cm}^{-2}\cdot\text{s}^{-1}$
j_s	local solute flux in the membrane, $\mu\text{mol}\cdot\text{cm}^{-2}\cdot\text{s}^{-1}$
J_w	volumetric water flux, $\text{cm}\cdot\text{s}^{-1}$
k	mass transfer coefficient, $\text{cm}\cdot\text{s}^{-1}$
L_p	hydraulic permeability coefficient, $\text{cm}\cdot\text{bar}^{-1}\cdot\text{s}^{-1}$
m_d	mass of the diafiltrate, g
m_f	mass of the feed solution in stirred cell, g
m_h	mass of solution in holdup, g
m_v	mass of the solution in scintillation vial, g
m_{tb}	mass of solution in the connecting tube, g
n	number of dissolved species, dimensionless
Pe	Peclet number, dimensionless
R	gas constant, $\text{cm}^3\cdot\text{bar}\cdot\mu\text{mol}^{-1}\cdot\text{K}^{-1}$
T	temperature, K
t_0	time stamp denotes applied pressure reaching target operating pressure, s
t_F	time stamp denotes the end of overflow, s
t_H	time stamp denotes that holdup volume is filled with permeate, s
t_L	time stamp denotes the end of lag, s
t_V	time stamp denotes that connecting tube is filled with permeate, s
v^0	average velocity within the stirred cell, $\text{cm}\cdot\text{s}^{-1}$
v_{head}	volume of the air in the headspace of the stirred cell, mL
$v_{tb,d}$	of air between the diafiltrate tank and the stirred cell, mL
v_{tb}	volume of solution in the tube between the stirred cell and scintillation vial, mL
$\Delta\mu$	chemical potential difference, $\text{J}\cdot\text{mol}^{-1}$
$\Delta\phi_D$	electrostatic potential difference, V
F	Faraday constant, $\text{C}\cdot\text{mol}^{-1}$
L	membrane thickness, cm
z	valency of the ion, dimensionless

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